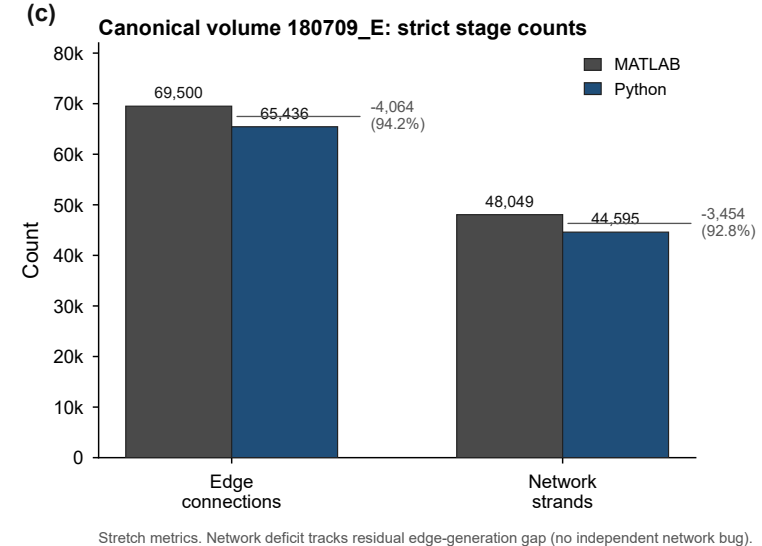
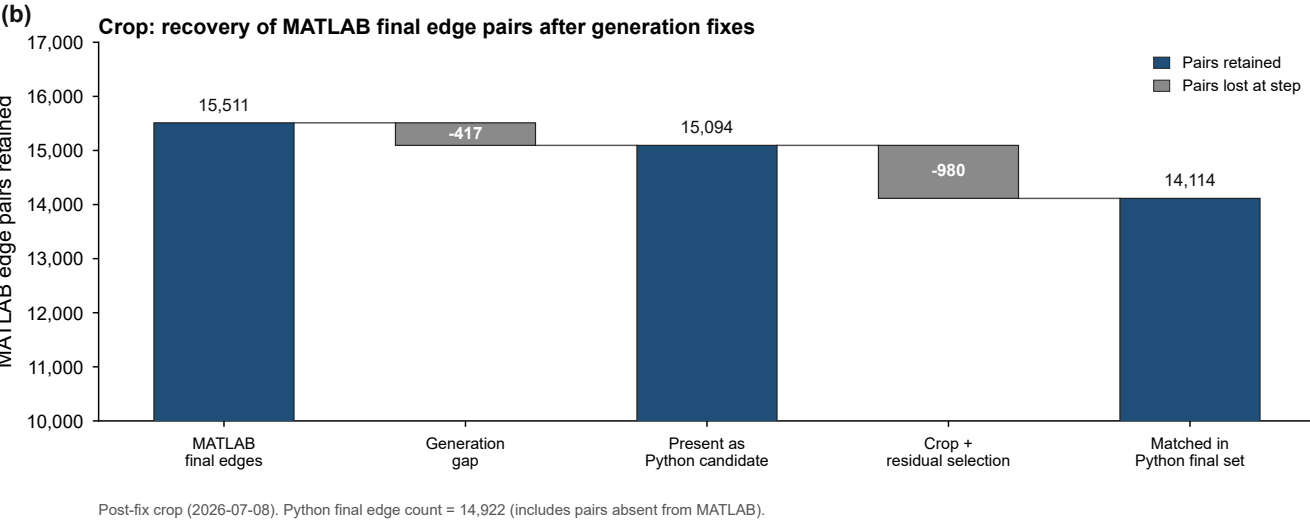
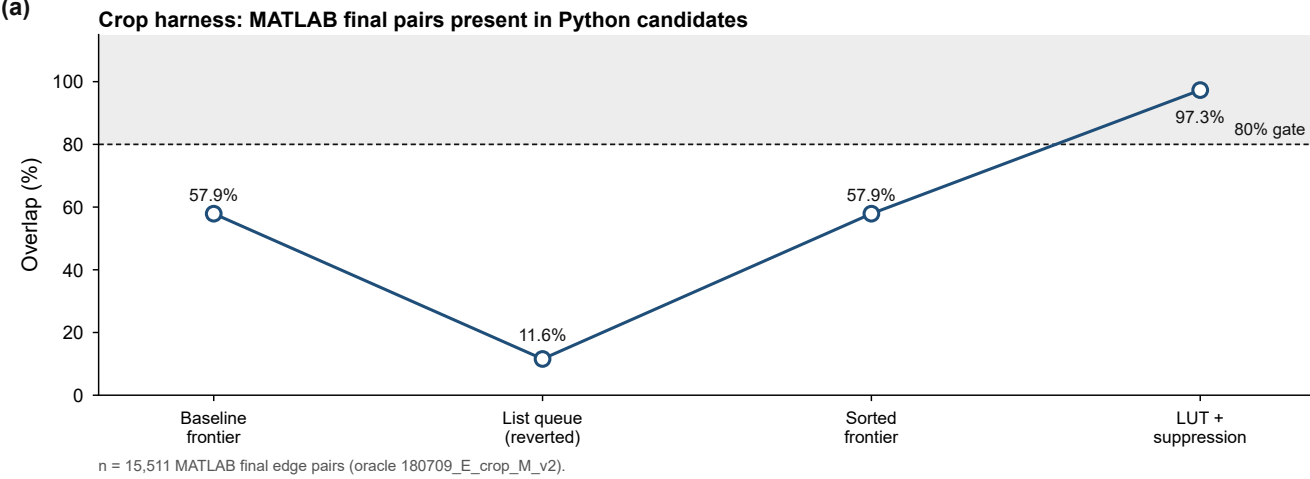




Certification metrics (same volume)

Stage	Metric	Result
Energy	scale-index mismatches	0 / 16.8M
Energy	max d float fields	~2e-11
Vertices	position / scale match	exact
Edges	ownership-map agreement	96.0%
Edges	ADR 0012 ownership bar	pass (>=60%)
Network	topology given MATLAB edges	exact*

*Isolated network with MATLAB edges matches topology exactly. Residual network fail is downstream of the edge-count gap only.